

Project Report:

Campus Waste Reduction Through Sustainable Material Substitution

in response to UVic Facilities Management RFP: EHSC202009

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University's Reduce Waste Goals

The main target for generating the literature review will be to compare the functions of reusable materials on campus with biodegradable materials.

Here are three goals in the 2020-2021 Sustainability Action Plan of University of Victoria to help to achieve:

Goal 1: Increase the waste diversion rate to 82% by Dec. 31, 2021.

Goal 2: Reduce the total amount of waste produced as measured in kilograms per campus user by 2021, relative to 2010 as the baseline year.

Goal 3: Incorporate waste reduction engagement in the Residence Services resident survey or educational programming.

Individuals could help to reduce waste in landfills by using rechargeable batteries, have food management, reduce junk mail, boycott plastics bags, etc...

University of Victoria also provides a Uvic Recyclepedia for reaching the goal, students can type the keyword on the search bar to find the destination of the waste.

The website of the Recyclepedia:

<https://www.uvic.ca/sustainability/topics/waste/index.php>

The Situation For Urban Canada's Landfill

Pollution is always a serious problem for every country. As **Figure 1** [1] showed, from 2002 to 2016, the total amount of solid waste collected in Canada increased by 3.5 million tonnes (or 11%). The amount of waste disposed of in landfills or incinerated increased by 0.9 million tonnes (or 4%) to reach 24.9 million tonnes in 2016. The amount of waste diverted grew by 2.6 million tonnes (or 39%) to reach 9.3 million tonnes in 2016. In 2016, the non-residential sector was responsible for 59% of disposed waste and 48% of diverted waste[1]. Landfills in Canada are regulated by provincial environmental agencies and environmental protection legislation. Older facilities tend to fall under current standards and are monitored for leaching. Some former locations have been converted to parkland. [2]

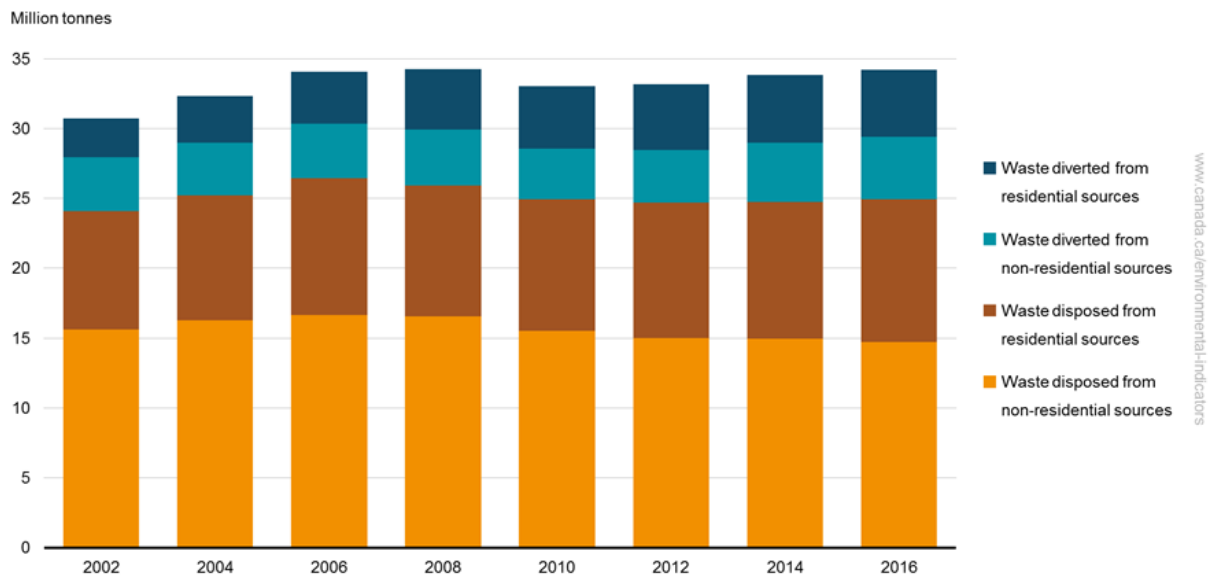


Figure 1. the total amount of solid waste collected in Canada from 2002 to 2016[1]

Changing Material to Reduce Waste to Landfill

Reducing the use of reusable materials is the most effective way for reaching goals, especially the food-related services. We propose to change most of the reusable materials' stuff that people use in their daily life, and materials that can be replaced can be selected as biodegradable material and recycling plastic.

Biodegradable material

Biodegradable plastics are plastics that can be decomposed by the action of living organisms, usually microbes, into water, carbon dioxide, and biomass. Biodegradable plastics are commonly produced with renewable raw materials, micro-organisms, petrochemicals, or combinations of all three[3]. As **Figure 2** [4] showed, the term biodegradable depicts a chemical process during which micro-organisms that are available in the environment convert materials into natural substances such as water, carbon dioxide and compost (artificial additives are not needed). The process of biodegradation depends on the surrounding environmental conditions (e.g. location or temperature), on the material and on the application. Compared to classical plastics, a big advantage of biodegradable plastics from renewable resources is the possibility of recollection and composting of material waste[4].

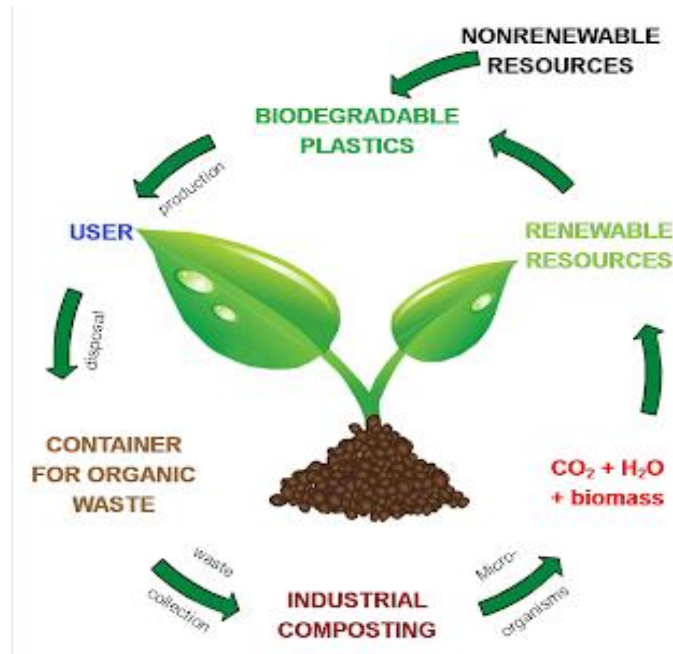


Figure 2. The circular process of biodegradable plastics. [4]

Recycling Plastic

- There is 4.6 million tonnes of plastic waste generated in Canada per year, which is roughly 135 Kg (kilograms) for each person. Based on the prediction by experts, the plastic waste number will reach around 6.0 million tonnes per year in 2030. [5]
- Rather than promoting recycled plastic tableware, to curb the use of recycled plastic works better depends on the new updated consumption between recycling plastic and directly producing plastic products.



[6]

From the line graph, we can see the blue line represents the cost of PET production per tonne, vice versa, the red line represents the cost of recycled PET per tonne. The PET production was usually higher than recycled plastic until around 2020. The gap has been narrowed down gradually from 2012 to 2019. By the year 2020, the consumption for PET has already been exceeded by recycled PET.

Waste will produce bad pollution on both air and water. Animals and marine lives face a greater threat due to the oil spills and leaching of chemicals. There are some poorly waste management systems:

1. Soil Contamination
2. Water Contamination
3. Climate contamination
4. Air contamination
5. Harm Towards Animal and Marine life
6. Human damage

Research Objectives

Our research goal has two key components:

- research
- student surveys

The main goal of research is to learn about landfill, the general problem we are facing, for example, pollution and solutions to fight for this problem. We also collect data from online research. If possible, we will like to interview the responsible staff for the uvic project. In addition, find a company that has already started a waste project and achieved great success.

Student surveys will give us a general idea about how people on campus could deal with the recycle problem. Many students come from different countries, so we can learn how other countries' perspectives toward recycling. Hence, we can collect a large amount of information about real life challenges on landfill.

Each part takes approximately two week. A week of researching information, interviewing and analyzing collections. We will need another week for student surveys and analyze data collected from the survey.

Project details	Week 1	Week 2	Week 3
research			
student survey			

Credentials

Xian Zhou is a mathematics and statistics department student at the University of Victoria. Xian owns effective data analysing skills especially for graphs.

Conclusion

If our research proves that replacing the reusable materials with biodegradable materials is practicable and the university agrees to implement it, we believe it can reach a close-to-zero percent of waste on our campus. Therefore, less disposed materials would dump to landfills, and a decreased amount of reusable materials is needed. Furthermore, we would suggest calling for an action for eliminating the use of materials on campus for a lower cost. This helps to save the budget for university, reduce the waste, and assist to reach a 0% waste sending.

References

A. Gross, "Surge in single-use PPE feeds 'toxic' pandemic waste crisis: Free to read," *Financial Times*, 23-Jul-2020. [Online]. Available: <https://www.ft.com/content/d5e27b1f-ba5a-4445-a329-1802cf70d619>. [Accessed: 18-Oct-2020].

"Biodegradable plastic," *Wikipedia*, 11-Oct-2020. [Online]. Available: https://en.wikipedia.org/wiki/Biodegradable_plastic. [Accessed: 18-Oct-2020]. [3]

"Biodegradable plastics - Indonesian Sustainable Plastic," *Google Sites*. [Online]. Available: <https://sites.google.com/site/indonesiansustainableplastic/bioplastics/biodegradable-plastics>. [Accessed: 18-Oct-2020]. [4]

"Campaign," *Oceana Canada*. [Online]. Available: https://oceana.ca/en/our-campaigns/plastics/campaign?gclid=CjwKCAjwz6_8BRBkEiwA3p02VUYY5wzSrwx6nCMAScFx2u3NEiMrNFII9VHeiP0mmcnzrqaeYII6rxoCFQEQA_vD_BwE. [Accessed: 18-Oct-2020].

E. and C. C. Canada, "Government of Canada," *Canada.ca*, 20-Dec-2018. [Online]. Available: <https://www.canada.ca/en/environment-climate-change/services/environmental-indicators/solid-waste-diversion-disposal.html>. [Accessed: 18-Oct-2020]. [1]

"Landfill," *Wikipedia*, 10-Oct-2020. [Online]. Available: <https://en.wikipedia.org/wiki/Landfill>. [Accessed: 18-Oct-2020]. [2]

"UVic's Waste reduction hub", *University of Victoria*, [Online].

Available:<https://www.uvic.ca/sustainability/topics/waste/index.php>. [Accessed: 18-Oct-2020].

"Simple way to Reduce Waste in Landfills", *permaculture research institute*, [Online].

Available:<https://www.permaculturenews.org/2017/03/27/simple-ways-reduce-waste-landfills/>. [Accessed: 18-Oct-2020]

"6 Negative Effects of Improper Waste Management" *MTS*, 08-Oct-2017. [Online].

Available:<https://www.metropolitantransferstation.com.au/blog/negative-effects-of-improper-waste-management>. [Accessed: 18-Oct-2020].